# Exit Ticket · reflect and retrieve

## Exit Ticket

A many-to-many relationship can always be represented correctly with just a single FK on one of the two entity tables.

## Use key vocabulary from memory

1. Explain entity in your own words.

2. Explain relationship in your own words.

3. Explain cardinality in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 135 · Entity Relationships: One-to-Many and Many-to-Many

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is the role of a foreign key?

### Q2

In Big O notation, what is binary search worst-case complexity?

### Q3

What does ROM stand for?

### Q4

Which is removed first from a FIFO queue?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Identify entities — Find objects with independent identity, not every noun.

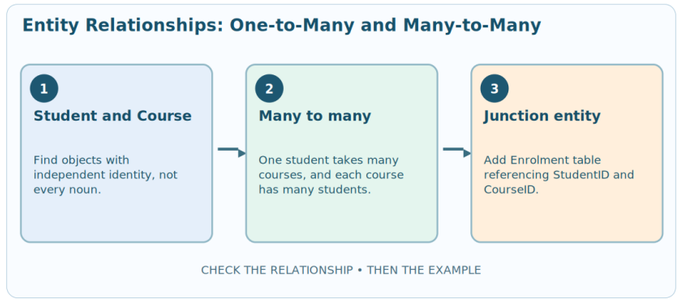
## Core idea 2

Set cardinality — One student takes many courses, and each course has many students.

## Core idea 3

Resolve M:N — Add Enrolment table referencing StudentID and CourseID.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Evaluate why a field containing comma-separated ClubIDs is poorer than a Membership junction table. [8]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Explain the difference between one-to-many and many-to-many relations with examples. [6]

## Question 2

Design tables for Author and Book when each book has multiple authors. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

entity — An identifiable class of thing stored in database.

relationship — Association between entities.

cardinality — How many related instances are possible.

junction table — Table breaking a many-to-many relation into two one-to-many relations.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Represent students enrolling on several courses, with many students per course.

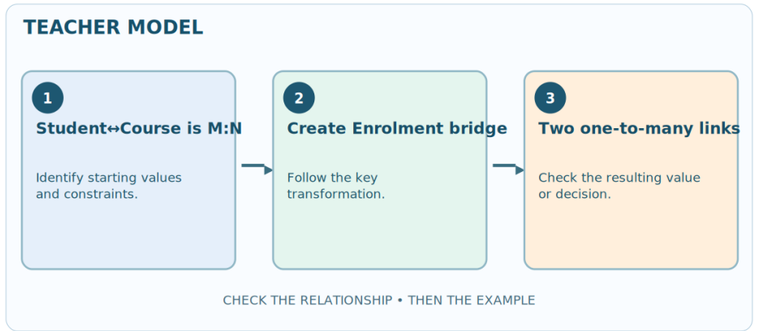
## Follow the worked method

1. Student↔Course is M:N — Find the given data and name the required result.

2. Create Enrolment bridge — Explain why this intermediate stage is necessary.

3. Two one-to-many links — Student(StudentID PK) and Course(CourseID PK) form many-to-many. Resolve with Enrolment(StudentID FK, CourseID FK, EnrolledOn), with composite PK (StudentID, CourseID) if enrolment is unique per pair.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

One Customer
Find the given data and name the required result.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.